CLOSED SETS ARE STABLE UNDER SPECIALIZATION, OPEN SETS UNDER GENERALIZATION, AND THE TWO STABILITIES ARE COMPLEMENTARY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The specialization relation between points and the definitions of stability under specialization and under generalization. Set up in the author's section on specialization, immediately before the counted statement. Fixes the three assertions and the relation used throughout.

Lemma 0.1. *Let X be a topological space.*

(1) *Any closed subset of X is stable under specialization.*
(2) *Any open subset of X is stable under generalization.*
(3) *A subset $T \subset X$ is stable under specialization if and only if the complement T^c is stable under generalization.*

Proof. Let F be a closed subset of X, if $y \in F$ then $\{y\} \subset F$, so $\overline{\{y\}} \subset \overline{F} = F$ as F is closed. Thus for all specialization x of y, we have $x \in F$.

Let $x, y \in X$ such that $x \in \overline{\{y\}}$ and let T be a subset of X. Saying that T is stable under specialization means that $y \in T$ implies $x \in T$ and reciprocally saying that T is stable under generalization means that $x \in T$ implies $y \in T$. Therefore (3) is proven using contraposition.

The second property follows from (1) and (3) by considering the complement. □

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